# AJAY VIKRAM PERIASAMI

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## Education

**Duke University** — North Carolina, USA
M.S. in Computer Science (AI/ML); GPA: 4.0/4.0 — Aug 2025 – May 2027

**National Institute of Technology Karnataka** — Surathkal, India
B.Tech. in Computer Science and Engineering; GPA: 9.46/10; Rank: 4/130 — Dec 2020 – Apr 2024

## Research Interests

Embodied intelligence, multimodal world models, hierarchical planning, control and failure recovery, representation learning.

## Research Experience

**The Robot Dexterity Lab, Duke University** — May 2026 – Present
Graduate Researcher; Advisor: Xianyi Cheng

- Developed **FRAMES**, a hierarchical framework for humanoid loco-manipulation integrating VLM skill planning, failure monitoring and recovery. Created perception skills for object grounding and memory module to improve skill selection.
- Developed a VLM monitoring module using temporal state-differences of multi-view observations, proprioception, and contact feedback, achieving 94% accuracy across 100 MuJoCo trials on 5 tasks. Currently improving the recovery module.

**DukeNLP, Duke University** — Sep 2025 – Present
Graduate Researcher; Advisor: Bhuwan Dhingra

- Developed **Vision2Code**, a reference-code-free image-to-code benchmark spanning 539 test examples, 15 datasets, and six domains. Our dataset-specific VLM rubrics beat generic rubrics and other visual similarity baselines on human validation.
- Benchmarked nine VLMs, analyzed failures and developed an evaluator-guided self-training pipeline using rejection sampling, improving Qwen3.5-9B by 16% and Phi-4-MM by 28%, with gains confirmed by two VLM judges.
- Training general-purpose user simulators while curating large-scale, multi-domain training data from 100+ sources.

**Duke Institute for Brain Sciences, Duke University** — Sep 2025 – Present
Graduate Research Assistant; Advisor: Andrew M. Michael

- Developed a 3D-UNet-based motion correction model for fMRI data without requiring motion regressors at inference, improving correlation with ground truth by 46.1%. Extending to more preprocessing steps and realistic motion simulation.
- Developed a JEPA-style self-supervised model for phenotype prediction and test-retest reliability. Beat previous methods on 6/10 phenotypes and achieved comparable reliability with 26x fewer parameters and the smallest embedding size.
- Characterized and studied gray- and white-matter fMRI using frequency, synchrony and entropy based measures.

**Rhodes Information Initiative, Duke University** — May – Jul 2026
Data+ Researcher; Advisor: Andrew M. Michael

- Investigated structure–function coupling in human brain using DTI-derived fractional anisotropy and fMRI measures.
- Trained regression models for behavior prediction using Partial Least Squares to study effect of coupling and feature importance. Created a toolbox to compute structural and functional measures with in depth analysis and comparisons.

**NeuRonICS Lab, Indian Institute of Science** — May 2024 – Jul 2025
Machine Learning Research Intern; Advisor: Chetan Singh Thakur

- Built asynchronous, real-time trackers for astronomical objects by clustering neuromorphic events. Optimized code for low-power inference on telescopes. Achieved 55% higher tracking speeds than prior work at 1.1 µs/event latency.
- Compressed CNNs and state-space models (SSMs) using quantization, ternarization, pruning, and mixed precision methods. Achieved 16× raw weight compression with under 1% accuracy loss on radar human activity recognition.
- Developed shift-based event-camera time surface preprocessing methods, reducing FPGA LUT utilization by 70%.
- Benchmarked SSMs like S4, Mamba, and LMU for gesture, speech emotion, and human activity recognition.
- Designed low-latency LSTM models for primate neural decoding, and S4 models for auditory attention decoding.

**Computer Science Department, National Institute of Technology Karnataka** — Oct 2023 – Mar 2024
Undergraduate Researcher; Advisor: Jeny Rajan

- Developed **ViTFuser** for self-driving, with multi-resolution ViT fusion of RGB and LiDAR features and a Feature Pyramid Network for multi-scale object detection, improving driving score by 26.8% over TransFuser with 67% fewer parameters.

## Publications

### Conference Papers

[1] **Asynchronous High-Speed Tracking of Astronomical Objects Using Neuromorphic Camera for Edge Computing.**
Satyapreet Singh Yadav, **Ajay Vikram P**, Adithya M D, Chandra Sekhar Seelamantula, Chetan Singh Thakur.
*IEEE ICASSP 2026 – Oral* [paper]

[2] **ViTFuser: Advancements in Global Context Understanding for Autonomous Vehicles.**
Atanu Chatterjee, **Ajay Vikram P**, Aniketh Narayan Bellala, T. Naga Tarun, Basavaraj Talawar, M. Vani, Jeny Rajan.
*MIND 2024* [paper]

**Workshop Papers**

[3] **FRAMES: Failure Recovery And Monitoring of Embodied Skills for Humanoid Loco-Manipulation.**
**Ajay Vikram P**, Xinyuan Luo, Haoyu Li, Xianyi Cheng.
*Full-Shift Robot Co-Workers Workshop, IROS 2026* [paper]

[4] **Reinforcement Learning for the Game of Balatro: A Fast Simulator and Two-Policy Training Approach.**
**Ajay Vikram P**, Anjaneya Gupta, Ethan Cheng, Ivan Rosales.
*Experimental AI in Games Workshop, AAAI AIIDE 2026* [paper]

**Under Review**

[5] **Vision2Code: A Multi-Domain Benchmark for Evaluating Image-to-Code Generation.**
**Ajay Vikram P**, Junlin Wang, Bhuwan Dhingra.
*ICLR 2027 – Under Review* [paper]

[6] **A Novel Deep Learning Approach to fMRI Head Motion Correction without Motion Regressors.**
**Ajay Vikram P**, Bharat Kashyap Karri, Andrew M. Michael.
*MIDL 2027 – Under Review*

[7] **Characterization of Resting-State fMRI in Gray and White Matter: Comparison of Signal Measures and Network Patterns.**
**Ajay Vikram P**, Bharat Kashyap Karri, Arnav Gupta, Tina Yuan, Leonard E. White, Andrew M. Michael.
*IEEE ISBI 2027 – Under Review*

[8] **HOMI: Ultra-Fast EdgeAI platform for Event Cameras.**
Shankaranarayanan H, Satyapreet Singh Yadav, Adithya Krishna, **Ajay Vikram P**, Mahesh Mehendale, Chetan Singh Thakur.
*IEEE Internet of Things Journal – Under Review* [paper]

[9] **Predicting Biological Sex from Resting-State Functional Connectivity in Early Adulthood: Effects of Tissue Type, Brain Atlas, and Brain Volume.**
Bharat Kashyap Karri, **Ajay Vikram P**, Leonard E. White, Andrew M. Michael.
*Human Brain Mapping – Under Review*

## Posters & Abstracts

**Learning Reliable and Behaviorally Informative Representations from Resting-State fMRI.**
ISMRM 2027 – Abstract (under review)

**FRAMES: Failure Recovery And Monitoring of Embodied Skills for Humanoid Loco-Manipulation.**
IROS 2026 Workshop on Full-Shift Robot Co-Workers – Poster presentation

**fMRI Measures of the Human Brain and Associations to Behavior.**
Duke Neurology Research Symposium, 2026 – Poster presentation

**Structure-Function Coupling in the Human Brain and Links to Behavior.**
Duke Neurology Research Symposium, 2026 – Poster presentation

**Comparison of DMN and other Brain Networks during resting fMRI.**
IEEE EMBC 2026 – Abstract (accepted)

**Neuromorphic Camera for Event-Based Motion Detection and Tracking.**
Indian Institute of Science Open Day, 2025 – Poster presentation and live demonstration

## Selected Research Projects

**Multi-Task LLM Post-Training** Feb – Apr 2026

- Developed a four-stage LoRA post-training pipeline for Llama-3.1-8B using SFT and GRPO, investigating task interference, data alignment, and reward sparsity. Achieved 72.3% across IFEval, GSM8K, and HumanEval, ranking 2nd/22 teams.

**Balatro RL** Feb – Apr 2026

- Built a high-throughput Rust simulator and Gymnasium RL environment for Balatro, training a two-policy PPO architecture in Ray RLlib for tactical play and strategic deckbuilding, reaching approximately ante 3 of 8 on average.

**SkillBlenderLC** Sep – Dec 2025

- Extended SkillBlender with LLM-based task decomposition, missing-skill detection, and reward generation for humanoid skill acquisition via PPO in Isaac Gym. Evaluated six LLMs and identified limitations in reward design.

**Event-Based Drone Tracking** Sep – Dec 2024

- Developed a non-neural drone tracker using event clustering and Hungarian matching; maintained the selected track in 198/200 windows over a 10-second segment and optimized the event processor in C for deployment on Raspberry Pi 4.

**Ternarizing CNNs for Human Activity Recognition** Jun – Aug 2024

- Developed a ternary CNN with 8-bit activations for radar-based activity recognition, combining quantization-aware training, 45% sparsity pruning, and integer-only inference to achieve 98.91% accuracy, within 0.51 points of FP32.

## Achievements

**3rd place & Most Inclusive Design Award, Neurotech BCI Hackathon at NC State** 2026
Contribution: EEG-based trigger-word and double-blink detection models.

**17th place, IEEE ICASSP EEG Auditory Attention Decoding Challenge** 2026
Contribution: efficient state-space models for auditory attention decoding.

**Dean's Scholarship, Boston University** 2025
Offered $20,000 for the M.S. in Artificial Intelligence.

**5th place, IEEE BioCAS Grand Challenge on Neural Decoding** 2024
Contribution: efficient LSTM models for primate neural decoding.

**Annual Proficiency Prize for Academic Excellence, Bishop Cotton Boys' School** 2011-2018
Seven consecutive awards for outstanding academic performance.

## Teaching

**Probabilistic Machine Learning, Duke University** Jan – Apr 2026
Graduate Teaching Assistant

- Led lab sessions, held office hours, mentored projects, and graded assignments. Received a student evaluation of 4.5/5.

## Industry Experience

**Lamarr** Mar – Apr 2024
Machine Learning Intern; Remote

- Fine-tuned Qwen1.5-14B for domain-specific applications and integrated Table Transformer and TrOCR into existing backend for document understanding; developed LLM agents for document preparation and automated client outreach.

**Qualcomm** May – Jul 2023
Software Intern; Hyderabad, India

- Implemented a diff engine to compare memory usage across builds, surfacing regressions and optimizations.
- Built and optimized a memory-analysis tool, cutting processing time by 85% and memory usage by 75%.
- Developed a React interface featuring interactive memory graphs, filtering, sorting, and tree-based memory organization.
- Automated end-to-end workflows and CI/CD using Jenkins, Docker, and Kubernetes, including load balancing, authentication and automated email notifications; received a full-time offer for the tool's successful adoption.

## Leadership & Service

**Member, Computer Vision Team, Duke Robotics Club** Aug – Dec 2025
Profiled and optimized object detection inference on NVIDIA Jetson Nano.

**Bangalore Neuromorphic Engineering Workshop, IISc** Jan 2025
Assisted with workshop organization.

**Placement Coordinator, Career Development Center, NITK** Aug 2023 – Apr 2024
Coordinated campus recruitment for full-time roles and internships across 20+ companies.

**Volunteer Mentor, Leadership Development, Team Everest NGO** Feb – Mar 2024
Mentored an underprivileged student 1:1 in leadership development skills.

## Technical Skills

**Programming:** Python, C/C++, MATLAB, Rust, JavaScript, SQL.
**Machine learning:** PyTorch, TensorFlow, scikit-learn, Tinker, Brevitas, Llama Factory.
**RL & robotics:** Gymnasium, Ray RLlib, MuJoCo, ROS 2, CARLA.
**Scientific Computing:** NumPy, SciPy, OpenCV, TorchIO, NiBabel, Nilearn.
**Systems & tooling:** Linux, Slurm, Git, Docker, Kubernetes, Jenkins, MLflow.